

K-Space Axiomatization — Registration-Logic Foundation for VVV-QMRF

Framework: VietVunVut Quantum Measurement Registration Framework (VVV-QMRF)

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Scope: Axiomatize the K-side registration space as a formal registration-logic structure. This document provides the mathematical/logical foundation that the working paper v2.0 structural definitions rest upon.

Out of scope: This document does not modify Standard Quantum Mechanics, does not change any VVV-QMRF postulate (E1-E16), does not upgrade claim classes of paper v2.0, and does not claim K-space is a canonical QM object.

DISCLAIMER / CNH BO: VVV-QMRF is independent Class D personal research, not Standard Quantum Mechanics, not peer-reviewed or experimentally validated, and not for real-world technical use.

Abstract

This document formalizes the K-side registration space of VVV-QMRF (VietVunVut Quantum Measurement Registration Framework) as a registration-logic structure. VVV-QMRF extends quantum mechanics at the registration layer by introducing a K-space K_R , separate from the physical Hilbert space H , whose elements are K-state tuples recording measurement-registration acts, outcomes, self-certification markers, timestamps, and validity statuses.

The axiomatization is organized in two layers. **Layer 1** provides the core axiomatic foundation: K1 defines the carrier set; K2 imposes a strict total temporal order within K_R ; K3 formalizes self-certifying registration; K4 establishes default validity; K5 governs invalidation via registered contradiction; K6 defines cross-registration authority; K7 formalizes registration closure; K8 guarantees validity-preservation through cross-space embeddings. **Layer 2** provides bridge theorems connecting K1K8 to the Level 4 structural definitions of VVV-QMRF Working Paper v2.0: T1 constructs the admissible joint K-space K_{joint} ; T2 proves AdmJoint as a joint-validity preservation condition; T3 proves the K_{joint} failure proposition; T4 establishes global commutativity of embeddings.

A concrete model (minimal Extended Wigner's Friend scenario: two observers, one registration event each) demonstrates K1K8 consistency and verifies the Level 4 derivation chain. Two declared gaps remain: a framework-level semantic commitment (relativization defense, G1) and a temporal dependency on Level 4 boundary clause freeze (G3). All claims are Class D (proposed).

Keywords: quantum measurement, registration framework, K-space axiomatization, Buddhist epistemology, Prama, Wigner's friend, VVV-QMRF, self-certification, validity propagation

0. RCA Motivation

0.1 Define Symptom vs. Cause

Symptom	Working paper v2.0 Section 7.2 lists "Axiomatize K as a full mathematical structure" as a deferred item. Three other deferred proof items are all blocked by the absence of axiomatized K-space.
Root cause	K is defined extensionally rather than intensionally (via axioms that determine the properties of the space). All operations on K are defined ad-hoc per use case rather than derived from an axiomatic structure.

0.2 Trace 5 Whys

- **Why needed?** VVV-QMRF uses K as a "space" in formal claims without mathematical foundation.
- **Why now?** Three deferred proof items in paper v2.0 are blocked by this absence.
- **Why not before?** Paper achieved Class C/D claims with structural definitions; operational bridges sufficed.
- **Why is this right timing?** Paper formal chain is complete (Section 7.2). Axiomatization now serves dual purpose: foundation for proof upgrades AND quality audit of the paper's formal chain before community feedback arrives.
- **Root cause:** K was introduced architecturally but never given formal axiomatic definition. This was intentional architectural debt to prioritize operational contact. Debt is now due.

0.3 Isolate The Gap

K is currently a **collection** without structure. To be a **space**, it requires at minimum:

- **Carrier set** what elements belong to K
- **Order structure** temporal ordering of registration events
- **Validity structure** how validity propagates through order
- **Operations** embedding (morphism between K-spaces), joint construction (for `K_joint`)

0.4 Fundamental Design Decision

K-space is NOT a pure mathematical space. It is a **registration-logic structure**: a mathematical carrier (poset with morphisms) whose primitive predicates are epistemological . This is not Hilbert space, not phase space, not probability space these are all (math + math). K-space is (math + registration-logic). The mathematical structure is the **carrier**, not the **content**.

0.5 2-Layer Architecture

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Layer 1  CORE AXIOMS (K1-K8): Frozen
        Based on dependency stack Level 0-3 .
        These do NOT depend on Level 4 .
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Layer 2 BRIDGE THEOREMS (T1-T4): Updatable
Connect core axioms to Level 4 structural definitions.
Marked "pending Level 4 freeze" updatable without changing K1-K8.

1. Core Axioms Layer 1 (Frozen) / Tin Li Tng 1

AXIOM K1 Carrier Set / Tp nn

Statement:

The K-side registration space of a registering system R is a set K_R whose elements are K-state tuples. Each tuple contains five fields: measurement-registration act identifier (M), registered outcome (o), self-certification marker (cert), registration time (t), and validity status (V).

Formal:

$K_R = \{ k \mid k = M, o, cert, t, V \}$

where:

- M M_K measurement-registration act identifier
- o $O \{ \}$ registered outcome
- cert $\{0,1\}$ self-certification marker (admission filter at K_R boundary)
- t T_R registration time (discrete index or real-valued timestamp)
- V $\{0,1\}$ validity status

Cert admission rule:

$k \in K_R \implies cert(k) = 1.$

Reason: K_R is "produced by R over time" every element of K_R is a registration event that has occurred. By K3, occurrence implies ${}_R(M) = 1$, hence $cert(k) = 1.$

The cert $\{0,1\}$ range is retained for the admission-filtering boundary:
events with $cert = 0$ are NOT admitted into K_R (admission failure outside K_R scope).

K_R is produced by registering system R over time.
 K_R is finite or countably infinite (discrete sequence of registration events).

Property	Value
Source	Level 3: K-state tuple from meta_architecture/vvv_qmrf_meta_architecture_registration_layer_formalization.md 1
BE lineage	Prama cognition as structured event: act , object (prameya), self-awareness , result (phala), validity
Claim class	C (conjectural VVV-QMRF formal definition)
Dependency	Level 3 (K-state tuple). No Level 4 dependency.
Boundary	K_R is not a Hilbert space, not a set of physical density matrices, not a probability space. Elements k are registration states they record what was registered, not what physically exists. The $o =$ slot is reserved for E9 (null event) and E14 (validated absence) but is not operationalized in this axiom set.
Consistency	K1 is consistent with K H (Level 1). The carrier set contains registration tuples, not physical states.

AXIOM K2 Temporal Order / Th t Thi gian

Statement:

$(K_R, <_R)$ is a strict total order (chain) where $k1 <_R k2$ iff $t(k1) < t(k2)$ for $k1, k2 \in K_R$ produced by the same registering process R . Within a single K_R , all registration events are comparable via their timestamps. The order is discrete: between any two consecutive registration events k_i and k_{i+1} , there is no K -side registration-state identity. **Clarification on order type:** Within a single K_R , the order is **total** (every pair of elements is comparable) because all elements share the same timestamp space T_R . The term "partial order" applies only when considering the **cross-K-space** setting: elements from different $K_R, K\{R\}$ with independent timestamp spaces are NOT comparable. K2 axiomatizes the intra- K_R case, which is a strict total order. The inter-K-space partial order emerges in T1 (K_joint construction) where independent timestamp spaces are combined.

Formal:

For all $k1, k2 \in K_R$:

$k1 <_R k2$ iff $t(k1) < t(k2)$

$(K_R, <_R)$ is a strict total order (chain):

(i) Irreflexive: $(k <_R k)$

(ii) Transitive: $k1 <_R k2 \wedge k2 <_R k3 \implies k1 <_R k3$

(iii) Asymmetry: $k1 <_R k2 \implies (k2 <_R k1)$ [follows from (i)+(ii)]

(iv) Totality: $k1, k2 \in K_R, k1 \neq k2 \implies k1 <_R k2 \vee k2 <_R k1$

[all elements comparable: $t(k1) < t(k2)$ for distinct events]

Note: Totality (iv) holds because distinct registration events in the same K_R have distinct timestamps. If two events were to share a timestamp, they would be the same registration event (identity by timestamp within K_R).

Discreteness :

For any consecutive pair $k_i, k_{i+1} \in K_R$ with no k' such that $k_i <_R k' <_R k_{i+1}$:
 $t(k_i) < t(k_{i+1})$, $\text{RegistrationState}(t) =$

Property	Value
Source	Level 2: E6 (Registering-System-as-Process) + S2- lemma
BE lineage	Kaabhagavda momentariness: registration time is discrete; no continuous registration identity between events
Claim class	D (proposed)
Dependency	Level 2 . No Level 4 dependency.
Boundary	Does NOT claim physical time is discrete. Only registration-time is discrete within K_R . Physical time in H remains continuous as per Standard QM. The "no registration-state identity between events" is a K -side property, not a claim about physical continuity.
Order type note	Within a single K_R , the order is strict total (chain), not merely partial. The partial-order structure emerges only in the cross-K-space setting (K_joint , T1). Previous versions mislabeled this as "strict partial order" corrected in v1.2.
Consistency	K2 is consistent with E6's definition of R as an ordered sequence $\{M_1, \dots, M_n\}$ with $t(M_1) < \dots < t(M_n)$. E6's total ordering of the sequence confirms strict totality. The discreteness clause directly instantiates the lemma.

AXIOM K3 Self-Certification / T chng nhn

Statement:

For each $k \in K_R$, the certification marker $\text{cert}(k) = _R(M) \in \{0,1\}$ is determined intrinsically within K_R , not by any external registration act $M' \vdash M$ and not by any registering system $R' \vdash R$. $_R(M) = 1$ iff M has occurred as a K-side registration event of R .

Formal:

$_R(M) = 1$ iff M has occurred as a K-side registration event of R ,
and this occurrence is determined intrinsically within K_R .

For all $k \in K_R$:

$\text{cert}(k) = _R(M)$ [cert tracks self-certification]

Observer-indexed independence:

$_R(M)$ is independent of $_R\{R\}$ for any $R \vdash R$.

$_R(M)$ does not require $M \vdash M$ such that $_M\{M\}(M) = 1$.

Reflexivity (E1 core property):

The certification of M 's occurrence is part of M 's own K-side instantiation.

No second-order meta-registration chain is required on the K-side.

Property	Value
Source	Level 2: E1 (Self-Certifying Registration)
BE lineage	Svasavedana self-awareness/self-certifying cognition: a cognition certifies its own occurrence without requiring a second-order cognition
Claim class	D (proposed)
Dependency	Level 2 (E1). No Level 4 dependency.
Boundary	$_R(M)$ certifies the OCCURRENCE of M as a K-side registration event. It does NOT certify that the physical outcome is true/correct. It is NOT consciousness, NOT a physical detector response, NOT a second-order measurement act. It is a structural property of K-side registration events the K-side analogue of "the detector clicked" being registered as "the detector clicked" without needing another detector to detect the first detector.
Consistency	K3 is consistent with E1's definition of (M) as intrinsic to M . The observer-indexed extension matches paper v2.0 3.3. K3 does not require the equivalence of (M) and R_svasa formalisms.

AXIOM K4 Default Validity / Tnh hp l Mc nh

Statement:

For any $k \in K_R$ with $\text{cert}(k) = 1$, the validity status $V(k) = 1$ upon instantiation of k in K_R . Validity is the default state of a self-certified registration event; it does not require external confirmation.

Formal:

For all $k \in K_R$:

$\text{cert}(k) = 1 \implies V(k) = 1$ (upon instantiation of k in K_R)

Exception E9 null registration event:

For $k_{\text{null}} \in K_R$ where:

$o(k_{\text{null}}) = \text{(no registered outcome)}$

$I(k_{\text{null}}) = 0$

$\text{cert}(k_{\text{null}}) = 1$ (self-certified: interaction occurred)

$V(k_{\text{null}}) = 0$ (by definition of null event, not by K5 invalidation)

This overrides the default rule K4. Null events are self-certified (interaction occurred) but carry zero outcome information $V = 0$.

Default rule:

For non-null k : $V(k)$ starts at 1. No external act is required to establish $V(k) = 1$.

Provision: $V(k) = 1$ is provisional until the registration process closes (per E7: V_{prov} vs V_{final} distinction; see K7 Closure Axiom).

Property	Value
Source	Level 2: E7 Axiom 1 (Default validity)
BE lineage	Svata prmya intrinsic validity: a cognition is valid by default in virtue of its occurrence
Claim class	D (proposed)
Dependency	Level 2 (E7 Axiom 1). No Level 4 dependency.
Boundary	$V(k) = 1$ is default K-side registration validity for non-null events. It does NOT mean the physical outcome is correct, does NOT mean the Born-rule probability was calculated correctly, and is NOT absolute metaphysical truth. It only means: within K_R , this registration event is treated as valid until contradicted. The E9 exception (null events: $\text{cert}=1, V=0$) is consistent with self-certification logic: $_R(M)$ certifies occurrence, not outcome validity. A null event certifies "interaction occurred" but its outcome is (zero information) $V=0$ by definition, not by K5 contradiction. The provisional/final distinction is formalized in K7 (Closure Axiom).
Consistency	K4 is consistent with E7 Axiom 1. K4 works with K3: only self-certified events ($\text{cert}=1$) get default validity. Events that fail admission ($\text{cert}=0$) have no validity status defined by this axiom. K4 + exception clause is compatible with E9: null events are self-certified (interaction occurred) but carry $V=0$ because outcome information is /zero the exception does not break the default rule for non-null events.

AXIOM K5 Invalidation / V hui ha

Statement:

$V(k_1) = 0$ iff there exists $k_2 \in K_R$ with $k_1 <_R k_2$ such that k_2 stands in registered contradiction to k_1 within a shared K-side comparison context C_K , and k_2 has valid cross-registration authority with respect to k_1 . Validity cannot be externally confirmed (only contradicted). Pre-closure: $V_{\text{prov}}(k_1) = 0$ is reversible in principle if the contradicting act is itself invalidated before process closure (K7). Post-closure: $V_{\text{final}}(k_1) = 0$ is irreversible and absolute.

Primitive predicate definitions:

• **Registered Contradiction minimal operational definition:**

For $k_1, k_2 \in K_R$ within a shared comparison context C_K : $k_2 \prec k_1$ holds iff the registration contents $o(k_1)$ and $o(k_2)$ cannot both be treated as valid K-side claims within the same C_K . This is a registration-layer contradiction, NOT a physical contradiction, NOT logical inconsistency in Standard QM, and NOT a claim that either physical event failed to occur. Minimal condition: $k_2 \prec k_1$ when: - k_1 registers a definite outcome $o(k_1)$ - k_2 registers a content in which `` is not preserved as a valid claim - The validity of both claims cannot be simultaneously maintained without contradiction. Full formalization of conditions (including boundary clauses: not physical erasure, not null event, not invalid when both sides are independently valid) is in Level 4. The above minimal definition is sufficient for K5 operational closure.

• **Comparison Context (C_K):**

A comparison context C_K is a K-side structure in which registration contents from multiple K-state tuples are evaluated for mutual consistency. C_K is defined by the D_joint demand: a C_K exists iff $requires_K_joint = 1$ for the relevant K-spaces.

• **Cross-Registration Authority deferred to K6:**

The "valid cross-registration authority" condition is formalized in Axiom K6 below.

Formal:

Firing precondition:

C_K must exist i.e., $requires_K_joint = 1$ for the relevant K-spaces.
K5 is a cross-observer invalidation rule: it fires only in joint-registration (K_joint) contexts. When $requires_K_joint = 0$, no C_K exists, condition (ii) is undefined, and K5 does not fire.

$V(k_1) = 0$ iff $k_2 \in K_R$ such that:

- (i) $k_1 \prec k_2$ [K_2 : k_2 is later in registration order]
- (ii) $k_2 \prec k_1$ within shared C_K [registered contradiction in comparison context]
- (iii) k_2 has valid cross-registration authority [authority conditions from paper v2.0 4.4] with respect to k_1 in C_K

Asymmetry (E7 Axiom 3):

F such that $F \models \{V(k) = 1\}$
(No external function can restore or confirm validity.)

Validity stages (K7):

$V_prov(k)$ provisional validity, during open registration process (pre-closure)
 $V_final(k)$ final validity, after registration process closes (post-closure)
K5 conditions (i)-(iii) govern V_prov transitions during the open process.
 $V(k)$ in K5 refers to $V_prov(k)$ pre-closure and $V_final(k)$ post-closure.

Irreversibility (post-closure only):

$V_final(k) = 0$ $V_final(k)$ remains 0 permanently.
(After registration process closes (K7), invalidation is absolute and cannot be revised.)

Pre-closure (K7):

$V_prov(k) = 0$ is in principle reversible: if the contradicting act k_2 is itself invalidated ($V(k_2) = 0$) before process closure (K7), the K5 trigger for k_1 is removed and $V_prov(k_1)$ is no longer forced to 0. Governed by K7 pre-closure K5 transitions are not final.

K_R disambiguation (cross-space context):

When C_K exists ($requires_K_joint = 1$), the quantifier $k_2 \in K_R$ operates over the relevant subspace of K_joint : k_2 may originate from a different K-space K_X and appears as $i_X(k_2) \in K_joint$. The concrete model uses $i_W(k_W) \in K_joint$ the operative reading of K_R is K_joint when C_K exists. In isolated scenarios

(requires_K_joint = 0, no C_K), K5 does not fire because condition (ii) is undefined .

Property	Value
Source	Level 2: E7 Axioms 2-3 (Invalidation + Asymmetry)
BE lineage	Parata prmya invalidity is detected extrinsically. Bdhaka prama a contradicting cognition retroactively voids the earlier cognition.
Claim class	D (proposed)
Dependency	Level 2 (E7 Axioms 2-3). Uses ` and "cross-registration authority" as primitive predicates whose full formalization is in Level 4 . K5 asserts the structural rule; the precise conditions for k2 k1 and "valid cross-registration authority" are defined in the bridge theorems (T1-T3). 2nd-order Layer 2 dependencies Dep-A: C_K existence precondition requires Level 4 predicate requires_K_joint=1 ; K5 does not fire when C_K is absent (also reflected in Layer 1 Summary K5 row Role 1, via F4). Dep-B (F7a non-circularity guard): K5 condition (i) is defined natively by K2's <_R ordering. In cross-space application, K5 can only be evaluated after a candidate K_joint has already been constructed. T1 constructs <_joint> from K2 native orders + Level 4 cross-structure temporal relations + K8 field preservation; K5 does not define or prove <_joint>. Therefore Dep-B is an application-order dependency only: K5 uses <_joint> inside K_joint` after T1 supplies the candidate joint order, so no circular dependency is introduced.
Boundary	K5 is a registration-layer invalidation rule. It does NOT claim that the physical outcome of M_1 is retroactively erased from -side history. The physical interaction I_1 still occurred; only its K-side registration validity is revised. The `` relation is NOT physical orthogonality in H. The irreversibility of V0 is a K-side property, not a claim about physical time asymmetry.
Consistency	K5 is consistent with E7 Axioms 2-3 and with the act-level contradiction definition in paper v2.0 4.4. The primitive predicates now have minimal operational definitions: is defined above in K5, C_K is tied to requires_K_joint, and authority is formalized in K6 below. This removes the prior dependency inversion where frozen K5 relied on undefined Level 4 primitives.

AXIOM K6 Cross-Registration Authority / Thm quyn Cho

Statement:

For $k_1, k_2 \in K_R$ with $k_1 \leq_R k_2$, k_2 has valid cross-registration authority with respect to k_1 in comparison context C_K iff: (a) both k_1 and k_2 belong to the same C_K , (b) $V(k_2) = 1$ at the time of the authority check (k_2 has not itself been invalidated), and (c) the validity demand D_{joint} that defines C_K includes the claim-content of k_1 within its scope. Cross-registration authority is NOT a hierarchy of observers; it is a structural relation within a shared comparison context.

Formal:

$\text{Auth} = 1$ iff all of:

- (a) $C_K\text{-sphere}(k_1) = C_K\text{-sphere}(k_2)$ [both belong to same C_K]
- (b) $V(k_2) = 1$ [k_2 not invalidated at check time]
- (c) $k_1 \in \text{scope}(D_{\text{joint}})$ [k_1 's claim falls within D_{joint} scope]

where:

$C_K\text{-sphere}(k)$ is the comparison context that k belongs to.

$\text{scope}(D_{\text{joint}})$ is the set of registration acts whose claims D_{joint} demands joint validity evaluation for.

Auth is NOT:

- Observer hierarchy (no "Wigner over Friend" privilege)
- Physical measurement authority
- Absolute epistemic authority (bound to C_K , not universal)
- Transitive across distinct C_K contexts:

$\text{Auth} \text{ Auth} \text{ Auth}$ [when $C_K \subset C_{K'}$]

Note: within a single shared C_K the formal block does not exclude transitivity.

Non-transitivity proof sketch (counterexample):

Consider three registration events $k_1, k_2, k_3 \in K_R$ with $k_1 \leq_R k_2 \leq_R k_3$, and TWO distinct comparison contexts C_K and $C_{K'}$:

Setup:

- $\text{Auth} = 1$: $k_1, k_2 \in C_K$, $V(k_2)=1$, $k_1 \in \text{scope}(D_{\text{joint}} \text{ of } C_K)$.
- $\text{Auth} = 1$: $k_2, k_3 \in C_{K'}$, $V(k_3)=1$, $k_2 \in \text{scope}(D_{\text{joint}} \text{ of } C_{K'})$.
- But $k_1 \notin C_{K'}$ (k_1 is not in the second comparison context).

Claim: $\text{Auth} = 0$ in BOTH contexts:

- In C_K : $k_3 \notin C_K$ condition (a) fails $\text{Auth} = 0$.
- In $C_{K'}$: $k_1 \notin C_{K'}$ condition (a) fails $\text{Auth} = 0$.

Therefore: $\text{Auth} \text{ Auth}$ holds, but

$\text{Auth} = 0$ in every available comparison context.

Scope: this counterexample proves non-transitivity across distinct C_K contexts. It does not claim non-transitivity within a single shared C_K .

Physical intuition: Authority is context-bound. Two different D_{joint} demands create two different comparison scopes. Transitivity would require a single C_K containing all three events with overlapping scope which is not guaranteed by the pairwise authority relations.

Property	Value
Source	Level 2: E7 Axiom 2 + paper v2.0 4.4 cross-registration authority section
BE lineage	Bdhaka prama a contradicting cognition must be a valid cognition itself to serve as bdhaka. An invalid cognition cannot void another.
Claim class	D (proposed)

Property	Value
Dependency	Level 2 (E7 Axiom 2). Uses C_K and D_joint from Level 4 with three Auth roles: (1) existential precondition K6 operates only when C_K exists (requires_K_joint = 1); (2) C_K-sphere membership parameter (condition a); (3) D_joint scope parameter (condition c). Dep-A (2nd-order Layer 2 dependency): Auth requires C_K to exist ; K6 does not operate when C_K is absent analog K5 Dep-A. Conditional semantic dependency (I-03 pattern): condition (c) $k1 \text{ scope}(D_joint)$ means Level 4 changes to D_joint scope alter which $k1$ have Auth = 1, without altering K6 text (syntactic freeze holds); analogous to K5_K boundary clauses (I-03/F3).
Boundary	Auth is a K-side structural relation within C_K. It does NOT create observer hierarchy, does NOT grant absolute epistemic privilege, and is NOT a claim about physical measurement authority. Two observers in the same C_K may have mutual authority (symmetric) when both are valid. Authority is lost the moment $V(k) \ 0$ (K5).
Consistency	K6 operationalizes K5's condition (iii) "valid cross-registration authority" is no longer an undefined primitive. K6 is consistent with E7 Axiom 2's extrinsic invalidation: a contradicting act must itself be valid. K6's non-transitivity preserves the pair-wise nature of authority checks.

AXIOM K7 — Registration Process Closure

Statement:

The registration process R of K_R closes at registration time t_{close} when no pending requires_K_joint demands remain for any pair of K-spaces involving K_R . At closure, for all $k \in K_R$: $V(k)$ transitions from provisional status $V_{prov}(k)$ to final status $V_{final}(k)$. After closure: (a) no new k can be instantiated in K_R , (b) K5 irreversibility becomes absolute ($V(k) \ 0$ cannot be revised by any future event), and (c) no new D_joint involving K_R can be raised. Before closure, all $V(k)$ are provisional and subject to K5 invalidation.

Formal:

R closes at $t_{close}(K_R)$ iff:
 pairs (K_R, K_X) where X is any registering system:
 $pending(K_R, K_X) =$

where $pending(K_R, K_X)$ is the set of unresolved requires_K_joint demands
 whose D_joint involves both K_R and K_X .

At closure:

For all $k \in K_R$:
 $V_{prov}(k) \rightarrow V_{final}(k)$ [provisional final]

Post-closure properties:

- (a) K_R is closed under new k : $k_{new} \in K_R$ with $t(k_{new}) > t_{close}$
- (b) K5 irreversibility is absolute: $V_{final}(k) = 0 \rightarrow V_{final}(k)$ stays 0 permanently
- (c) No new D_joint can be raised
- (d) K_joint involving K_R becomes final (no reconfiguration)

Pre-closure:

All $V(k)$ are $V_{\text{prov}}(k)$. K5 invalidation transitions modify V_{prov} .

The V_{final} value for each k is the limit of $V_{\text{prov}}(k)$ as $t \rightarrow t_{\text{close}}$.

Property	Value
Source	Level 2: E7
BE lineage	Nicaya (ascertainment/determination) a cognition becomes determinate when the cognitive process reaches closure; before that, it is provisional
Claim class	D (proposed)
Dependency	Level 2 (E7 $V_{\text{prov}}/V_{\text{final}}$ distinction). Uses requires_K_joint and D_{joint} from Level 4 as closure condition inputs. Dep-B (2nd-order Layer 2 dependency): K7 closure condition $\text{pending}(K_R, K_X)$ = uses the concept of "resolved demand" a requires_K_joint demand is resolved when a K_{joint} registration event satisfying T2 AdmJoint conditions has occurred; without T2, "resolved" is an undefined primitive in K7; T2 is a silent Layer 2 dependency for closure semantics analog K5 Dep-B ($T1 \leq_{\text{joint}}$ for K5 condition (i)). Conditional semantic dependency (I-03 pattern): Level 4 extensional scope of requires_K_joint (which event types require joint registration) directly determines t_{close} timing expanding the scope delays closure, narrowing it advances closure; K7 text is frozen but t_{close} depends on Level 4 content; analogous to K5 K boundary clauses (I-03/F3) and K6 D_{joint} scope (I-03/F6a).
Boundary	K7 defines when the K-side registration process closes. It does NOT claim the physical interaction has ended, does NOT claim the H-space state has reached a final value, and does NOT claim that all possible measurements have been completed. Closure is a K-side property: no more joint validity demands are pending. In practice, t_{close} may be identified with the end of an experimental protocol, the publication of results, or any agreed-upon endpoint of the registration process.
Consistency	K7 resolves the "provisional" gap in K4 and K5. K4's default validity is provisional until closure; K5's irreversibility is provisional until closure, then absolute. K7 is consistent with E7's distinction between V_{prov} and V_{final} . K7 does not conflict with K5: pre-closure K5 transitions are reversible in principle (if the contradicting act is itself invalidated before closure), but post-closure K5 transitions are absolute.

AXIOM K8 Cross-Space Embedding Preservation / Bo ton qua Php nhng

Statement:

For any embedding $i_{\{RX\}}: K_R \rightarrow K_X$ that maps a K-state tuple $k \in K_R$ into a joint or cross K-space K_X , the embedding preserves the validity value $V(k)$ at the moment of embedding: $V_X = V_R(k)$ at embedding time t_{embed} . Furthermore, the embedding preserves the remaining tuple fields $(M, o,$

cert, t) unchanged. After embedding, $V_X(k)$ evolves independently within K_X according to K4-K7 the embedding preserves the initial validity snapshot, not immunity from future invalidation.

Formal:

For any embedding $i: K_R \rightarrow K_X$ (where K_X may be a joint K-space K_{joint} or any cross-space structure):

(i) V-preservation (Embedding Postulate):

$V_X(i(k)) = V_R(k)$ at t_{embed} for all $k \in K_R$

(V-value carries over into K_X at the moment of embedding.)

(ii) Field preservation:

$M_X(i(k)) = M_R(k)$ [act identifier preserved]

$o_X(i(k)) = o_R(k)$ [registered outcome preserved]

$\text{cert}_X(i(k)) = \text{cert}_R(k)$ [self-certification marker preserved]

$t_X(i(k)) = t_R(k)$ [registration timestamp preserved]

(iii) Post-embedding evolution:

After t_{embed} , $V_X(i(k))$ evolves according to K4-K7 rules within K_X . K8 does NOT immunize against future K5 invalidation in K_X . It only guarantees that the embedding itself does not alter V the embedded element enters K_X with its native validity intact, then stands or falls by K_X 's own validity dynamics.

(iv) Non-redundancy:

K8 is NOT derivable from K4. K4 governs V upon instantiation within a native K_R it is silent on cross-space embedding behavior. K8 is an independent postulate about the behavior of V under the embedding operation. A framework without K8 (or an equivalent postulate) cannot guarantee that embedded registration acts retain their validity status.

Property	Value
Source	Architectural necessity identified during T1 construction: K1-K7 alone cannot guarantee V-preservation through cross-space embeddings because K4 only governs V upon instantiation within a native K_R . Formerly tracked as "Embedding Postulate (EP)" an external postulate required by T1. Promoted to K8 (v1.4) to make the axiom set self-contained.
BE lineage	Anugama (continuity/attendant relation) the validity of a cognition accompanies it when the cognition is taken up in a broader cognitive context; the cognition does not lose its epistemic status merely by being embedded in a larger framework
Claim class	D (proposed)
Dependency	Level 0-3 (K1 tuple structure, K4 default validity). No Level 4 dependency.
Boundary	K8 is a structural postulate about the embedding operation itself, not about the native K-spaces being embedded. It guarantees snapshot preservation, not permanent immunity. After embedding, K5 can still fire in K_X and invalidate $i(k)$. K8 does NOT claim that V -values are absolutely invariant under all operations only that the embedding map i itself does not alter them.

Property	Value
Consistency	K8 is consistent with K4: K4 defines validity at native instantiation, K8 defines validity-preservation at cross-space transfer. They govern different moments. K8 is consistent with K5: K8 guarantees the initial V snapshot in K_X, but K5 can still transition V_X(i(k)) 0 afterwards the embedded element inherits both rights (default validity) and liabilities (future invalidation) of its new home. K8 + K1 together guarantee that the embedded tuple maintains its identity (5-field structure) across spaces.

Layer 1 Summary / Tng kt Tng 1

Axiom	Content	Fields covered	Source level	Freeze status	Level dependency 4
K1	Carrier set K_R is a set of 5-field tuples	M, o, cert, t, V	Level 3	Frozen	None
K2	Temporal order strict total order (chain) within K_R, discrete	t (ordering)	Level 2	Frozen	None
K3	Self-certification _R(M) intrinsic to R	cert	Level 2	Frozen	None
K4	Default validity V=1 on instantiation (with E9 exception)	V (default)	Level 2	Frozen	None
K5	Invalidation V0 by later with authority; minimal definition included	V (transition)	Level 2	Frozen	C_K roles: (1) existential precondition K5 fires only when C_K exists (requires_K_joint = 1); (2) evaluation parameter (condition ii); (3) Auth evaluation parameter (condition iii)

Axiom	Content	Fields covered	Source level	Freeze status	Level dependency 4
K6	Cross-registration authority structural relation within C_K, non-hierarchical	V (authority condition)	Level 2	Frozen	C_K roles: (1) existential precondition for all Auth checks; (2) C_K-sphere membership parameter (condition a); (3) D_joint scope parameter (condition c)
K7	Registration process closure V_prov V_final, absolute irreversibility	V (closure)	Level 2	Frozen	Uses requires_K_joint for pending check only
K8	Cross-space embedding preservation V preserved at embedding time; fields preserved	V (embedding) + M, o, cert, t	Level 3	Frozen	None

Dependency isolation: K1-K8 depend ONLY on Level 0-3 . Where K5-K7 reference Level 4 concepts (C_K, D_joint, requires_K_joint), they reference them for **scope identification only** (e.g., "is k1 in the same C_K as k2?"), not for their internal structure or definition.

Syntactic freeze (unconditional): K1-K8 text is frozen. If paper v2.0 community feedback changes the internal structure of AdmJoint, K1-K8 do not change AdmJoint appears only in bridge theorems T1-T4, not in any K1-K8 axiom text.

Semantic dependency for _K (conditional): K5 minimal definition provides K5-local operational closure. However, Level 4 4.4 boundary clauses ("not null event", "not invalid when both sides independently valid") narrow K5 minimal and constitute a real semantic dependency: if these boundary clauses change, K5 fires in a different set of cases even though K5 text is unchanged. The syntactic freeze guarantee holds unconditionally; the semantic behavior guarantee for K5 is conditional on Level 4 boundary clauses remaining a conservative extension of K5 minimal (adding scope, not contradicting it). Only bridge theorems T1-T3 need updating for structural Level 4 changes.

2. Bridge Theorems — Layer 2 (Updatable)

Status note: Theorems T1-T3 are **pending Level 4 freeze** (paper v2.0 in community review). They connect K1-K8 to the paper's structural definitions. If Level 4 definitions change, T1-T3 are updated independently of K1-K8. T4 is new (Class D).

T1 K_joint Construction Theorem

Statement:

Given K -side spaces K_A and K_B of registering systems A, B : if $\text{requires_K_joint}(A, B) = 1$ via a shared validity demand D_{joint} , then a candidate joint K -space $K_{\text{joint}}(A, B)$ exists as the minimal K -space containing order-preserving embeddings $i_A: K_A \rightarrow K_{\text{joint}}$ and $i_B: K_B \rightarrow K_{\text{joint}}$ that preserve cert and V values. The embedding respects the internal time-order of each structure, and the combined order in K_{joint} is the transitive closure of the two embedded orders plus cross-structure temporal relations from the shared laboratory history.

Derivation from axioms:

$K_{\text{joint}}(A, B)$ exists as a candidate K -space:

K1: K_A, K_B are sets of tuples K_{joint} carrier = $i_A(K_A) \cup i_B(K_B)$
 + K2: each has temporal order combined order = $(i_A(<_A) \cup i_B(<_B) \cup \text{cross-rel})^{^+}$
 where $^{^+}$ is transitive closure
 + K3: embeddings i_A, i_B preserve cert values maps to same cert in K_{joint}
 + K8: embeddings i_A, i_B preserve V values at embedding time
 ($V_X(i(k)) = V_R(k)$ at t_{embed} native validity carries into K_{joint}).
 K8 replaces the former external Embedding Postulate (EP); the axiom set is now self-contained for V -preservation.
 + requires $K_{\text{joint}} = 1$ via D_{joint}
 + K6: cross-registration authority between embedded elements is evaluated within the C_K defined by D_{joint} .

Note: existence of a candidate K_{joint} does NOT guarantee it is admissible. Admissibility requires AdmJoint conditions (i)-(v) to hold .

F7a non-circularity guard:

T1 does not depend on K5 Dep-B. T1 constructs the candidate $<_{\text{joint}}$ order from K2 native orders, cross-structure temporal relations supplied by the Level 4 D_{joint} context, and K8 field/ V preservation under embedding. K5 is evaluated only after the candidate K_{joint} exists. Thus the dependency direction is:
 $K2/K8 + \text{Level 4 } D_{\text{joint}} \rightarrow \text{T1 candidate } K_{\text{joint}}/<_{\text{joint}} \rightarrow \text{K5 application inside } K_{\text{joint}}$.

Property	Value
Level 4 dependency	requires_K_joint predicate, D_{joint} definition . K6 authority structure.
Layer 1 dependency	K8 (cross-space embedding preservation) V -preservation is now derived from a core axiom, not an external postulate. Former EP gap (G1) resolved by promoting EP K8 (v1.4).
Claim class	D (proposed)
Freeze status	Pending Level 4 freeze
Update trigger	If requires_K_joint definition changes or AdmJoint embedding conditions change

T2

Statement:

For K -side spaces K_A and K_B : $K_A \cup K_B$ holds iff $\text{requires_K_joint}(A, B) = 1$ AND no admissible K_{joint} exists satisfying AdmJoint conditions (i)-(v) while preserving K4 (default validity) and K5 (no invalidation) for both embedded sides simultaneously. The incommensurability is traced to a K5 conflict: any candidate K_{joint} forces $V(k_A) = 0$ or $V(k_B) = 0$ while both are claimed as jointly valid.

Derivation from axioms:

$K_A _K K_B$

$\text{requires_K_joint}(A,B) = 1$ [Level 4, 4.3]

$K_joint: \text{AdmJoint}(K_joint; K_A, K_B) = 1$ [Level 4, 4.3]

T2 focuses on the case where AdmJoint fails via K5 conflict:

Under candidate K_joint , $k_A \ i_A(K_A)$, $k_B \ i_B(K_B)$ such that

$k_B \ k_A$ within C_K (registered contradiction) [K5 primitive, Level 4 4.4]

AND $\text{Auth} = 1$ (cross-registration authority) [K6]

K5 forces $V_prov(k_A) \ 0$ OR $V_prov(k_B) \ 0$ [K5 pre-closure]

AdmJoint condition (iv) violated [Level 4, 4.3]

(no V_prov invalidation while both are claimed jointly valid during the candidate K_joint admissibility check)

F7b timing guard:

T2's $\text{AdmJoint}(\text{iv})$ check is pre-closure. It operates on V_prov because K7 assigns V_final only after pending requires_K_joint demands have been resolved. Therefore T2 cannot use V_final as the admissibility input without creating a timing inversion between T2 and K7.

Note: K5 conflict is a SUFFICIENT condition for AdmJoint failure, NOT a necessary condition. AdmJoint may fail for other reasons (e.g., embedding structure fails conditions (i)-(iii) or (v); D_joint is ill-posed for the given K-spaces; K7 closure has already locked one K-space preventing reconfiguration). In such cases, $_K$ may still hold, but the derivation trace differs from the T2 K5-conflict path shown above.

CIRCULARITY ACKNOWLEDGMENT:

T2 derives $_K$ (incommensurability) from K5 conflict.

K5 uses (registered contradiction) as a primitive predicate.

The MINIMAL operational definition of $_K$ is given in K5 (Layer 1).

The FULL formalization of conditions is in Level 4 ,

which is itself defined in terms of "registered contradiction"

a concept whose precise boundary clauses are NOT yet frozen.

Status of the circularity:

- K5's minimal definition (Layer 1) is self-contained: it defines $_K$ as "registration contents cannot both be treated as valid K-side claims within the same C_K ." This is NOT circular.
- v1.3 CONCRETE MODEL FINDING: Circularity does NOT arise in the concrete model K5 minimal is directly verifiable by content inspection without invoking Level 4 full .
- The circularity only appears in the GENERAL case where T2 needs AdmJoint conditions that reference full and even there it is a dependency-ordering issue (Level 4 not frozen), not a logical circle.
- Resolution path: freeze Level 4 boundary clauses independently of T2, then T2's derivation becomes non-circular.
- Until Level 4 freeze: T2 is CONDITIONAL on Level 4 definition. T2's conclusion is valid IF the Level 4 formalization is consistent with K5's minimal definition. This is a reasonable assumption but NOT a proven fact.

Boundary clauses :

$_K$ does NOT assert that either physical event fails to occur on the -side.

$_K$ does NOT mean either observer's outcome is false within its own K-side.

$_K$ is NOT equivalent to $\text{Null_K}(e)$ null registration is separate (E9).

$_K$ applies only when both sides are valid/provisionally valid within their own K-side.

F7b K7 resolution semantics:

For K7 closure, a requires_K_joint demand is resolved when T2 yields one of two admissibility outcomes:

(1) Success path:

K_joint such that $AdmJoint(K_joint; K_A, K_B) = 1$.

The demand is resolved by successful joint registration.

(2) Failure path:

K_joint such that $AdmJoint(K_joint; K_A, K_B) = 1$.

The demand is resolved as $K_A _K K_B$.

In both cases, the pending demand is no longer open. K7 may then evaluate whether $pending(K_R, K_X) = .$ T2 supplies the resolution semantics; K7 performs the closure transition from V_prov to V_final .

Property	Value
Level 4 dependency	$AdmJoint$ conditions (i)-(v), $_K$ boundary clauses, D_joint
Layer 1 dependency	K5 (invalidation rule) + K6 (authority condition) + K8 (V-preservation at embedding) T2 derivation uses all three. K8 resolves former EP gap (G1): V-preservation is now derived from a core axiom.
Claim class	D (proposed). Derivation trace is Class D; the $_K$ conclusion matches paper v2.0 Class D definition.
Important	K5 conflict is a SUFFICIENT condition for $AdmJoint$ failure, not necessary. $AdmJoint$ may fail for other reasons. F7b: $AdmJoint(iv)$ is checked against V_prov during pre-closure admissibility testing; V_final is assigned only after K7 closure. T2 also supplies K7's "resolved demand" semantics: a demand is resolved either by successful $AdmJoint = 1$ or by failure $AdmJoint = 0$ producing $_K$.
Freeze status	Pending Level 4 freeze
Update trigger	If $AdmJoint$ conditions (i)-(v) change, or $_K$ boundary clauses are revised

T3 Bridge_EWF Formalization Theorem

Statement:

In an Extended Wigner's Friend configuration where D_joint requires F-side and W-side registrations to support one cross-observer validity constraint, $Bridge_EWF(D_joint; M_F, M_W) = 1$ is derivable from K5 when: (a) M_F registers a definite friend-side outcome o_F , (b) M_W registers the same laboratory as a coherent superposition in which no definite o_F is preserved as a W-side valid claim, and (c) no reinterpretation inside the same K_joint can preserve both registered contents without changing the validity claim of at least one side.

Derivation from axioms:

$Bridge_EWF(D_joint; M_F, M_W) = 1$

D_joint requires F-side and W-side registrations to be evaluated

as jointly valid parts of one laboratory registration history [Level 4, 4.3]

M_F : $k_F = M_F, o_F, 1, t_F, 1$ (definite outcome, self-certified, valid)

M_W : $k_W = M_W, o_W, 1, t_W, 1$ (superposition registered, no definite o_F)

Under candidate K_joint :

$k_W \ k_F$ within C_K (registration contents incompatible) [Level 4, 4.4]

k_W has valid cross-registration authority [Level 4, 4.4]

K5: $V(k_F) \neq \emptyset \text{ OR } V(k_W) \neq \emptyset$ [K5]
AdmJoint condition (iv) violated [Level 4, 4.3]
 $M_W \neq M_F$ (act-level registered contradiction) [Level 4, 4.4]

External semantic assumption Relativization defense:
T3 depends on the following framework-level semantic boundary
(not derived from K1-K8):
"If K_{joint} only hosts meta-descriptions ,
it does not satisfy D_{joint} (which demands joint validity of original claims).
Relativizing contents abandons D_{joint} rather than satisfying it."

This is a FRAMEWORK-LEVEL SEMANTIC COMMITMENT, not a theorem derivable from K1-K8. It defines what counts as "satisfying D_{joint} " inside this formulation of VVV-QMRF. If this assumption is rejected (i.e., if meta-descriptions ARE accepted as satisfying D_{joint}), then T3's conclusion ($Bridge_EWF = 1$) does not follow from K1-K8 alone.

Property	Value
Level 4 dependency	Bridge_EWF lemma, D_{joint} , cross-registration authority
Layer 1 dependency	K5 (invalidation) + K6 (authority) + K8 (V-preservation at embedding) T3 derivation uses all three.
External assumption	Relativization defense a framework-level semantic commitment, not derivable from K1-K8. T3 is conditional on this assumption. This is not an internal contradiction; it is a declared semantic boundary for what counts as satisfying D_{joint} in this formulation.
Claim class	D/C boundary
Freeze status	Pending Level 4 freeze
Update trigger	If Bridge_EWF sufficient conditions change, or cross-registration authority criteria are revised, or relativization defense is challenged

T4 N-Observer Generalization Theorem

Statement:

For $N \geq 2$ registering systems $R_1, \dots, R_N$ with K -side spaces $K_1, \dots, K_N$: the joint K -space $K_{joint}(R_1, \dots, R_N)$ exists as the colimit of the embedding diagram only when (1) every pair (i, j) with $requires_K_{joint}(K_i, K_j) = 1$ satisfies pairwise AdmJoint, and (2) the N -observer embedding diagram satisfies global overlap compatibility: shared K -state images and all embedding paths commute in the candidate K_{joint} . K -side incommensurability $_K$ is NOT necessarily transitive: $K_A _K K_B _K K_C$ does NOT entail $K_A _K K_C$. Each pair requires an independent D_{joint} and AdmJoint check, but pairwise admissibility alone does not prove N -way colimit commutativity.

Derivation from axioms:

$K_{joint}(R_1, \dots, R_N)$ = colimit of embedding diagram D where:
objects: $K_1, K_2, \dots, K_N$
morphisms: for each pair (i, j) with $requires_K_{joint}(K_i, K_j) = 1$
and AdmJoint satisfied: embedding $i_{\{ij\}}: K_i \rightarrow K_j$
colimit universal property: K_{joint} is the minimal K -space receiving embeddings from all K_i that commute with the diagram morphisms.

F7d global commutativity guard:

Pairwise `AdmJoint` is necessary but not sufficient for N-observer colimit existence. K8 preserves tuple fields and V values along each embedding, but K8 does not by itself guarantee path-independence across multiple embeddings. For $N > 2$, T4 requires an additional global compatibility condition: whenever two embedding paths carry the same source K-state or shared overlap into `K_joint`, their images must agree on `M`, `o`, `cert`, `t`, and initial V at embedding time, and their post-embedding validity dynamics must not force inconsistent identifications. Thus the diagram must commute globally, not merely pairwise.

Non-transitivity of `_K`:

Counter-example possibility:

`K_A _K K_B K_B _K K_C`

when: `requires_K_joint(A,C) = 0` (no shared validity demand)

OR: `requires_K_joint(A,C) = 1 AND AdmJoint(K_joint; K_A, K_C) = 1`

Each pair (i,j) must pass independent:

(a) `requires_K_joint` check (is joint validity demanded?)

(b) `AdmJoint` check (can joint validity be preserved?)

Number of pairwise checks for N observers:

At most $N(N-1)/2$ pairs with `requires_K_joint = 1` (when all pairs require joint check).

Not all pairs necessarily require `K_joint` (some may be causally/comparison-isolated).

Property	Value
Level 4 dependency	All Level 4 definitions, generalized to N observers; global overlap/path-commutativity condition for N-observer diagrams
Claim class	D (proposed) NEW. Not in paper v2.0 (which handles N=2 only).
Important	Pairwise <code>AdmJoint</code> checks are necessary local conditions, not sufficient global conditions. N-observer <code>K_joint</code> requires pairwise admissibility plus global overlap/path-commutativity. K8 supplies field/V preservation along each embedding but does not by itself prove global commutativity.
Freeze status	New theorem. Requires independent verification for $N > 2$.
Update trigger	When $N > 2$ EWF scenarios are modeled; when paper v3.0 extends to multi-observer cases

Layer 2 Summary / Tng kt Tng 2

Theorem	Bridges axioms to	Level 4 dependency	Freeze status	Risk if Level 4 changes
T1	<code>K_joint</code> construction	<code>requires_K_joint</code> , <code>D_joint</code> , embeddings	Pending	Theorem statement updates; K1-K8 unchanged
T2	<code>_K</code> derivation	<code>AdmJoint (i)-(v)</code> , <code>_K</code> boundary clauses	Pending	Derivation chain updates; K1-K8 unchanged

Theorem	Bridges axioms to	Level 4 dependency	Freeze status	Risk if Level 4 changes
T3	Bridge_EWF formalization	Bridge_EWF lemma, framework-level semantic boundary: relativization defense	Pending	Derivation chain + semantic boundary may need revision; K1-K8 unchanged
T4	N-observer generalization	All Level 4, generalized to N	New Class D	New theorem; independently updatable

3. Audit Matrices / Ma trn Kim ton

3.1 E1-E7 Core Postulate Audit

Question for each postulate: Are K1-K8 sufficient to capture its K-side structural content, or do the axioms contradict it?

Postulate	Content	K-space coverage	Verdict
E1	Self-Certifying Registration: $(M)=1$ intrinsic to M ; no M' required	K3 directly instantiates $_R(M)$ with intrinsic determination and observer-indexed independence	COVERED K3
E2	Registration Self-Completion: $M \wedge^K r$ (act-result inseparability)	Not directly axiomatized as a separate equivalence relation. K1 encodes E2 structurally: each admitted K-state tuple $k = M, o, cert, t, V$ co-instantiates the registration act M with its registered result o (r on the E2 side) in one K-side event, so no separate result-producing act is required. K4/K7 govern the tuple's validity lifecycle after instantiation; they support completion status but do not define the act-result inseparability itself.	ENCODED K1 tuple structure; K4/K7 validity lifecycle noted
E3	Registration Lock: $C: HK, C(l)=k_locked$	Not directly axiomatized. C is a bridge map, not an intra-K-space property. K1-K8 describe K-space structure; C belongs to the bridge layer.	OUT-OF-SCOPE Bridge layer. No conflict.

Postulate	Content	K-space coverage	Verdict
E4	Pre-Symbolic Registration Stratum: (M) K _{pre} , Sym=	Not directly axiomatized. K1 defines K-state tuples at the symbolic level (o is a symbolic outcome). The pre-symbolic stratum K _{pre} is a substructure not formalized in K1-K8.	OUT-OF-SCOPE Reserved for K-space stratification extension
E5	Internal Representation Encoding: f _{enc} maps apparatus state to outcome within K	Not directly axiomatized. f _{enc} is an encoding map that operates within K but is not a structural property of K-space itself.	OUT-OF-SCOPE Encoding operation. No conflict.
E6	Registering-System-as-Process: R = {M ₁ ,...,M _n }, no identity beyond acts	K2 directly instantiates the temporal order as a strict partial order. K1+K2 together encode: R IS the ordered set of K-state tuples there is no "R" separate from its K _R .	COVERED K1+K2
E7	Registration Validity Location: V=1 default (Axiom 1), V0 by (Axiom 2), asymmetry (Axiom 3)	K4 = Axiom 1 (default, with E9 exception). K5 = Axiom 2+3 (invalidation + asymmetry + irreversibility). K6 = authority condition. K7 = closure . All three E7 axioms + provisional/final distinction are directly instantiated.	COVERED K4+K5+K6+K7

E1-E7 Audit verdict: 3/7 COVERED directly (E1, E6, E7). 1/7 ENCODED implicitly (E2). 3/7 OUT-OF-SCOPE (E3, E4, E5). Zero contradictions. Coverage gaps are intentional .

3.2 E8-E16 Extension Postulate Audit

Question for each postulate: Does the postulate require K-space structure beyond K1-K8? If yes, is the gap documented?

Postulate	Content	K-space requirement	Verdict
E8	Retroactive Registration Override: M_2 retroactively voids M_1	K5 covers single-step invalidation: a later contradictory registration k2 can force $V_{prov}(k1) = 0$ before closure. K7 makes this pre-closure transition revisable: if the contradicting act k2 is itself invalidated before t_close, the K5 trigger for k1 is removed and $V_{prov}(k1)$ is no longer forced to 0. In cross-space cases where E8 is evaluated inside a requires_K_joint / C_K context, T2 supplies the resolved-demand semantics needed before K7 closure: the demand resolves either by successful AdmJoint = 1 or by AdmJoint = 0 producing _K. This E8 invalidation path is distinct from E9 null status: E8 is $V_{prov}0$ by K5 contradiction/authority, not definitional $V=0$ from $o=$. E8's orthogonality trigger condition remains a -side condition, not a K-space axiom. Multi-step retroactive chains (k3 voids k2, which re-opens k1) are still not fully formalized.	PARTIAL K5 single-step $V_{prov}0$ + K7 pre-closure re-assessment covered; T2 supplies resolved-demand semantics when E8 occurs in C_K; multi-step retroactive chain formalization deferred.
E9	Null Registering-System Event: interaction occurred but I=0	K1 reserves $o=$ slot. K4 includes explicit E9 exception clause: null events have cert=1 (interaction occurred) and $V=0$ by definition because zero outcome information is transferred. This $V=0$ is definitional null status, not K5/K6 invalidation: it requires no ` , no Auth, and no shared C_K. Therefore F2's K6 non-transitivity refinement across distinct C_K` contexts does not affect E9.	COVERED K1 $o=$ + K4 E9 exception; independent of K6/Auth/C_K.

Postulate	Content	K-space requirement	Verdict
E10	Tripartite Registration Validity Matrix: three validity criteria	Validity criteria operate on K-side predicates; K4-K5-K6-K7 provide the underlying validity structure. The tripartite matrix is a taxonomy layer on top of K4-K7.	COVERED K4-K7 as foundation. No new axiom needed.
E11	Contrapositive Quantum Evidence: evidence from absence	Evidence structure is outside K-space (bridge/evidence layer). Evidence ABOUT K-side states is not a property OF K-space.	OUT-OF-SCOPE Bridge/evidence layer. No conflict.
E12	Limit-Faculty Registration: different registering capacities	Different K_R types with different registration capacities are type-level distinctions, not new axioms. K1-K8 apply to all K_R regardless of capacity type.	COVERED Taxonomy layer. No new axiom needed.
E13	Temporal Discontinuity Registration	K2 already encodes discreteness via the lemma clause (no registration-state identity between events).	COVERED K2
E14	Validated Absence Registration: registration from absence of detection	Requires k_absence K_R with cert=1, o= (or o = "absence of X"), V=1 (valid absence). K1 reserves o= slot. K4's default validity applies ; the absence registration is not null (it carries positive information "X is absent"). The validity conditions for absence (expectation of detection + validated non-occurrence) are beyond K4-K5 scope.	PARTIAL K1 o= + K4 default validity structurally accommodate. Specific validity conditions for absence deferred.
E15	Intrinsic Relational Binding: entanglement as K-side relation	Relations BETWEEN K-spaces (K_A and K_B correlated via shared quantum state) are not covered by K1-K8, which are primarily intra-K-space axioms. T1 (K_joint) handles embeddings but not the nature of the binding relation itself.	GAP Inter-K-space relation structure not axiomatized. Reserved for K-space relation extension.

Postulate	Content	K-space requirement	Verdict
E16	Pre-Measurement Registration Indeterminacy: K-side state before first registration	K1-K7 describe K_R as a set of K-state tuples produced over time. The state BEFORE the first registration event (k_0 or pre-registration K-state) is not defined.	GAP Pre-registration K-state not defined. Reserved for K0 (pre-registration axiom).

E8-E16 Audit verdict: 6/9 COVERED or structurally accommodated (E9, E10, E11, E12, E13; E8 partial; E14 partial). 2 gaps (E15, E16). All gaps explicitly documented no hidden incompatibilities.

3.3 Operational Bridge Preservation Audit

Question for each bridge: Do K1-K8 invalidate or alter any operational bridge defined in paper v2.0?

Bridge	Paper	What it does	Preservation check	Verdict
Condition A	4.3	Wigner interference requires_K_joint=1	K1-K8 do not reference requires_K_joint directly. Bridge operates at Level 4 (D_joint). K-space axioms do not force or prevent requires_K_joint=1.	PASS Bridge unchanged.
Condition B	4.3	Direct comparison requires_K_joint=1	Same as above. K1-K5 are silent on comparison architecture.	PASS Bridge unchanged.
Condition B2	4.3	LF constraint requires_K_joint=1	Same as above.	PASS Bridge unchanged.
Condition C	4.3	No interference requires_K_joint=0	K1-K8 do not force K_joint construction. K_R remains isolated unless D_joint demands otherwise.	PASS Bridge unchanged.
Condition D	4.3	Separable state requires_K_joint=0	K1-K8 do not reference entanglement or separability.	PASS Bridge unchanged.
Condition E	4.3	Independent bookkeeping requires_K_joint=0	K1-K8 do not conflate K_R set membership with joint validity demands.	PASS Bridge unchanged.

Bridge	Paper	What it does	Preservation check	Verdict
ODC_K	4.6	Model-fit test for K _{joint} existence	K1-K8 define K-space structure but do not pre-determine ODC_K outcome. remains a free parameter. K4-K7 define validity propagation ODC_K tests whether a joint model preserving K4-K7 fits data.	PASS ODC_K unchanged. K4-K7 provide the validity constraints ODC_K checks.

Operational bridge audit verdict: 7/7 bridges preserved (no bridge broken by K1-K7). However: bridges B, B2, and ODC_K have an indirect semantic dependency on K4-K7 validity structure. K4-K7 define the validity propagation rules that these bridges operationalize. If K4-K7 were to change significantly, the semantic content of these bridges would shift even though their formal predicates (requires_K_{joint}, D_{joint}, AdmJoint) remain syntactically unchanged. This is a semantic dependency, not a syntactic break.

3.4 BE Source Lineage Audit

Question for each axiom: Is the axiom consistent with its BE structural source?

Axiom	BE source	BE claim	K-space instantiation	Consistency
K1	Prama (cognition as structured event)	A cognition has: act, object (prameya), self-awareness , result (phala)	K-state tuple has: M (act), o (object/outcome), cert (self-awareness marker), t (temporal index), V (validity/result status)	Consistent 5-field tuple maps onto prama structure
K2	Kaabhagavda (momentariness)	Cognition is momentary; no enduring cognitive substance between moments	Registration time is discrete; no K-side identity between consecutive events	Consistent discrete order matches momentariness without claiming physical time is discrete
K3	Svasavedana (self-awareness)	A cognition is self-aware; it illuminates both object and itself without a second cognition	_R(M) determined intrinsically within K_R; no M' required	Consistent intrinsic certification matches self-awareness
K4	Svata prmya (intrinsic validity)	Validity is intrinsic to cognition; it is the default, not something added by verification	V(k)=1 upon instantiation; no external act required to establish validity	Consistent default validity matches intrinsic validity

Axiom	BE source	BE claim	K-space instantiation	Consistency
K5	Parata prmya + Bdhaka prama	Invalidity is detected extrinsically; a contradicting later cognition voids the earlier one	V(k)0 only by later k' with and authority; asymmetry: no external function restores V=1	Consistent extrinsic invalidation matches bdhaka structure; asymmetry matches parata
K6	Bdhaka prama	A contradicting cognition must itself be valid to serve as a defeater; an invalid cognition cannot void another cognition	Cross-registration authority requires a valid later registration within the relevant shared C_K; invalid or out-of-scope registrations cannot invalidate k1	Consistent authority condition preserves bdhaka validity requirement
K7	Nicaya (ascertainment/determination)	Cognition becomes determinate when the cognitive process reaches closure; before closure, doubt or revision remains possible	Closure converts V_prov to V_final only after pending requires_K_joint demands are resolved	Consistent closure formalizes ascertainment without denying provisional pre-closure status
K8	Anugama (continuity/attendant relation)	A cognition retains its epistemic status when taken up in a broader cognitive context	Embedding preserves M, o, cert, t, and initial V at embedding time, while still allowing later validity dynamics	Consistent embedding preserves epistemic continuity without claiming immunity from later invalidation

BE lineage audit verdict: 8/8 PASS. Zero inconsistencies between K-space axioms and BE structural sources. Each core axiom K1-K8 preserves the "structural extraction, not identity" boundary.

4. Six-Condition Test Derivation from Axioms / Kim tra Su iu kin

Question: Can the six conditions for valid registered measurement be expressed in terms of K1-K8?

Condition	Original formulation	K-space expression	Derivable?
C1 (Physical)	X occurs at -side	Not a K-space condition. C1 is -side outside K1-K8 scope.	N/A -side condition
C2 (Admission)	X admitted into K-side as M_X for R	k K_R with M = M_X. Admission = instantiation of k in K_R. By K1 cert admission rule: cert(k)=1 for all k K_R.	K1: k K_R, cert(k)=1
C3 (Process membership)	M_X R where R = {M_R1, M_R2, ...}	k K_R, t(k) in the temporal order of K_R.	K1 + K2: k K_R with t(k) ordered

Condition	Original formulation	K-space expression	Derivable?
C4 (Self-certification)	$_R(M_X) = 1$, determined intrinsically	$\text{cert}(k) = _R(M_X) = 1$, determined within K_R .	K3: $\text{cert}(k) = _R(M)$
C5 (Default validity)	$V(M_X) = 1$ by default	$V(k) = 1$ upon instantiation .	K4: $\text{cert}=1 \ V=1$ (with E9 exception)
C6 (Non-invalidity)	No later M' contradicts M_X with authority	No $k' > k$ with $k' \leq k$ and $\text{Auth}=1 \ V(k)$ stays 1. Pre-closure: provisional. Post-closure (K7): final.	K5 + K6 + K7

Six-condition test verdict: 5/5 K-side conditions derivable from K1-K8. C1 is -side correctly outside K-space scope.

5. Claim Traceability / Truy vt Claim

Claim ID	Claim	Claim type	Source	Confidence	Boundary
C-KAXIOM-001	K_R is a set of 5-field K-state tuples (K1)	Class C formal definition	This document 1, K1	High	Not a Hilbert space; not a physical state space
C-KAXIOM-002	$(K_R, <_R)$ is a strict total order (chain) with discrete registration-time (K2)	Class proposed D	This document 1, K2; E6; S2-	High	Registration-time only; not physical time. Total within K_R ; partial only in cross-K-space (K_joint).
C-KAXIOM-003	$_R(M)$ is determined intrinsically within K_R (K3)	Class proposed D	This document 1, K3; E1	High	Certifies occurrence, not truth of outcome
C-KAXIOM-004	$V(k)=1$ by default for self-certified events (K4)	Class proposed D	This document 1, K4; E7 Axiom 1	High	Default K-side validity, not absolute truth
C-KAXIOM-005	$V(k)0$ iff later contradicting act with authority (K5)	Class proposed D	This document 1, K5; E7 Axioms 2-3	High	Registration-layer only; not physical erasure
C-KAXIOM-006	K_joint exists as colimit of embedding diagram (T1)	Class proposed D	This document 2, T1; paper v2.0 4.3	Medium pending Level 4 freeze	Candidate K_joint , not guaranteed admissible
C-KAXIOM-007	$_K$ derivable from K1-K5 + AdmJoint failure (T2)	Class proposed D	This document 2, T2; paper v2.0 4.4	Medium pending Level 4 freeze	Registration-layer incommensurability only

Claim ID	Claim	Claim type	Source	Confidence	Boundary
C-KAXIOM-008	Bridge_EWF derivable from K5 + EWF config (T3)	Class boundary D/C	This document 2, T3; paper v2.0 4.5	Medium pending Level 4 freeze	EWf-specific; not general LF theorem
C-KAXIOM-009	N-observer joint K-space is colimit; $_K$ non-transitive (T4)	Class proposed NEW D	This document 2, T4	Low unverified new, for N>2	Generalization; requires independent verification
C-KAXIOM-006a	K6: Auth=1 iff shared C_K , $V(k2)=1$, $k1scope(D_joint)$; non-hierarchical, non-transitive (K6)	Class proposed D	This document 1, K6; E7 Axiom 2; paper v2.0 4.4	Medium	Structural relation within C_K ; not observer hierarchy
C-KAXIOM-007a	K7: R closes at t_close when no pending requires $_K_joint$; V_provV_final ; post-closure irreversibility absolute (K7)	Class proposed D	This document 1, K7; E7 V_prov/V_final ; paper v2.0 2.2	Medium	K-side closure only; not physical process termination

Claim ID	Claim	Claim type	Source	Confidence	Boundary
C-KAXIOM-010	2-layer architecture isolates K1-K8 from Level 4 changes in two senses. (1) Syntactic isolation (unconditional): K1-K8 axiom text is frozen Level 4 changes (AdmJoint criteria, D_joint definitions, requires_K_joint scope) do not alter K1-K8 text. (2) Conditional semantic dependencies (K5/K6/K7): K5_K evaluation is narrowed by Level 4 boundary clauses (F3); K6 Auth evaluation depends on D_joint extensional scope (F6a); K7 t_close timing depends on requires_K_joint extensional scope (F6b). K1-K4 and K8 reference Level 4 for scope identification only or not at all. See 0.5 for full distinction.	Architectural claim	This document 0.5	High structural property	Architectural design, not mathematical theorem. Syntactic isolation is unconditional; semantic dependencies are conditional on Level 4 extensional content.

6. Non-Overclaim Guardrails / Ranh gii Chng Khng nh Qu mc

- **K-space is NOT a Hilbert space.** K_R is a set of registration tuples, not a vector space with inner product. K H is the core architectural commitment.
- **K-space axioms do NOT modify Standard QM.** P1-P4, Born rule, Schrdinger equation, and -side dynamics are unchanged.

- **K-space axioms are NOT physical laws.** They are proposed registration-layer structural definitions (Class D). They do not make empirically testable predictions independent of the operational bridges in paper v2.0.
- **K-space is registration-logic, not pure mathematics.** The axioms include primitive epistemological predicates that have no analogue in standard mathematical spaces. This is intentional K-space is a different kind of structure than Hilbert space.
- **Bridge theorems T1-T3 are pending Level 4 freeze.** They derive current paper v2.0 definitions from axioms. If community feedback changes those definitions, T1-T3 are updated K1-K8 are not. T3 additionally depends on an external philosophical assumption not derivable from K1-K8.
- **K1-K8 cover E1-E7, E9, E10, E11, E12, E13.** E8 (multi-step retroactive chain), E14 (validated absence beyond structural accommodation), E15, E16 require extensions deferred to future work. This is explicitly documented in 3.2.
- **This document does NOT upgrade any paper v2.0 claim class.** All claims remain Class D/C as in the paper. Axiomatization provides the foundation for future upgrades but does not perform them.
- **BE sources are structural lineage, NOT proof.** Each axiom annotates its BE source for traceability. The BE source is a structural analogue, not empirical evidence for the axiom's truth.

7. Concrete Model & Proof Attempt Level 4 Freeze Check / M hnh C th & Nhp Chng minh

Methodology: Following the "smallest model first, consistency before derivability" protocol. This section:

- Defines the smallest concrete EWF model (2 observers, 1 registration event each)
- Walks K1-K8 for consistency
- Walks Level 4 definitions for consistency
- Presents a proof attempt for T2 with gaps explicitly marked

7.1 Concrete Model Definition

Scenario: Extended Wigner's Friend (EWF), minimal configuration.

- **Friend F** measures spin of particle S inside sealed laboratory. Outcome: spin-up .
- **Wigner W** performs interference measurement on F's entire laboratory. Registers superposition $|+ =$. No definite o_F preserved as W-side valid claim.

Concrete K-spaces:

$K_F = \{ k_F \}$ where $k_F = M_F, |h, 1, t_F, 1$

M_F = "Friend measures spin of S"
 o_F = $|h$ (definite outcome: spin-up)
 $cert = 1$ (self-certified)
 $t_F = 1$ (registration time index)
 $V = 1$ (valid by default)

$K_W = \{ k_W \}$ where $k_W = M_W, |+ , 1, t_W, 1$

M_W = "Wigner interference measurement on F+S laboratory"

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o_W = |+ (superposition: no definite spin-up preserved)
cert = 1 (self-certified)
t_W = 2 (registration time index; after t_F in laboratory history)
V = 1 (valid by default)

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Model properties:

- $|K_F| = 1, |K_W| = 1$
- F and W are distinct registering systems
- Both are non-null events

7.2 K1-K8 Consistency Walk / Kim tra Nht qun K1-K8

Axiom	Check on K_F	Check on K_W	Result		
K1 (Carrier)	$k_F = M_F, \setminus$	$h, 1, 1, 1$ is a 5-field tuple. k_F K_F . $\text{cert}(k_F) = 1$ admitted.	$k_W = M_W, \setminus$	$+, 1, 2, 1$ is a 5-field tuple. k_W K_W . $\text{cert}(k_W) = 1$ admitted.	Both satisfy K1
K2 (Total order)	$K_F = \{k_F\}$: singleton. Trivially a strict total order (no pair to compare). Discrete: trivially satisfied.	$K_W = \{k_W\}$: singleton. Same reasoning.	Both satisfy K2		
K3 (Self-cert)	$_F(M_F) = 1$, determined within K_F . No M' M_F required. No $R' R_F$ involved.	$_W(M_W) = 1$, determined within K_W . No M' M_W required. No $R' R_W$ involved. $_F$ and $_W$ are independent.	Both satisfy K3		
K4 (Default V)	$\text{cert}(k_F) = 1$ $V(k_F) = 1$ upon instantiation. k_F is non-null. No E9 exception applies.	$\text{cert}(k_W) = 1$ $V(k_W) = 1$ upon instantiation. k_W is non-null. No E9 exception applies.	Both satisfy K4		
K5 (Invalidation)	No $k' K_F$ with $k_F <_F k'$. K_F has only one element. No invalidation possible within K_F . $V(k_F)$ remains 1.	No $k' K_W$ with $k_W <_W k'$. K_W has only one element. No invalidation possible within K_W . $V(k_W)$ remains 1.	K5 vacuously satisfied (no later event exists in either K-space)		
K6 (Authority)	No pair within K_F to check authority. Vacuously satisfied.	No pair within K_W to check authority. Vacuously satisfied.	Vacuously satisfied		

Axiom	Check on K_F	Check on K_W	Result		
K7 (Closure)	No pending requires_K_joint involving K_F K_F closes at $t_{\text{close}} = t_F$. $V_{\text{prov}}(k_F) = 1$ $V_{\text{final}}(k_F) = 1$.	No pending requires_K_joint involving K_W K_W closes at $t_{\text{close}} = t_W$. $V_{\text{prov}}(k_W) = 1$ $V_{\text{final}}(k_W) = 1$. BUT: see 7.3 requires_K_joint may be pending, preventing closure.	Conditional depends on D_joint status (Level 4)		
K8 (Embedding)	Intra-K-space: k_F has no embedding to check (K_F is native). Vacuously satisfied.	Intra-K-space: k_W has no embedding to check (K_W is native). Vacuously satisfied. Tested in K_joint context at L4-7 below.	Vacuously satisfied intra-K-space; tested cross-space in 7.3		

K1-K8 intra-K-space consistency verdict:

K_F and K_W each individually satisfy K1-K8 without contradiction. K5, K6, and K8 are vacuously satisfied because each K-space has only one element. K7 is conditionally satisfied: if requires_K_joint = 1 (Level 4), then closure is blocked until D_joint is resolved, making V_final pending. This is not an inconsistency it is K7 working as designed. K8's embedding preservation is tested in the cross-space K_joint construction at L4-7.

7.3 Level 4 Definitions Walk / Kim tra nh ngha Tng 4

Walking through each Level 4 definition applied to the concrete model.

Step L4-1: requires_K_joint predicate

`requires_K_joint(F, W) = ?`

Check conditions :

- (a) K_F and K_W are each valid within their own K-side? YES
- (b) Are they brought under a shared validity demand D_joint? YES EWF setup demands both be assessed as parts of one laboratory registration history
- (c) Does D_joint require both to be parts of the same YES LF/no-go constraint requires registration target/history/validity claim? F's and W's outcomes to be assigned simultaneous cross-observer validity
- (d) Can D_joint be evaluated while leaving K_F, K_W NO the comparison demands in fully independent K-spaces? embedding into one candidate K_joint
- (e) Does preserving D_joint require a candidate K_joint? YES

`requires_K_joint(F, W) = 1` via Condition A (Wigner interference)

Check	Status
Condition A (Wigner interference)	W performs interference on F+S lab. M_W registers superposition. M_F registers definite outcome. Both concern same lab history.

Step L4-2: D_joint predicate

$D_joint(K_F, K_W, Arch_EWF) = 1$

Arch_EWF = "Extended Wigner's Friend: F measures S inside lab;
W performs interference on F+S; LF comparison demands
both claims support one cross-observer validity constraint."

D_joint evaluates to 1 because Arch_EWF demands that K_F and K_W
support one shared registration-validity claim about the same laboratory.

Check	Status
D_joint = 1	Consistent with paper v2.0 4.3 definition

Step L4-3: Comparison context C_K

C_K exists for (k_F, k_W)?

Check conditions :

- (a) Both acts admitted into same comparison domain? YES D_joint demands it
- (b) Both indexed to same registration target/history? YES same laboratory F+S
- (c) Comparison does not presuppose both already YES comparison TESTS whether jointly valid? they can be jointly valid

C_K(k_F, k_W) exists.

Check	Status
C_K exists	All three conditions met

Step L4-4: Cross-registration authority

Auth = ?

Check conditions :

- (a) C_K-sphere(k_F) = C_K-sphere(k_W)? YES both in same C_K (L4-3)
- (b) $V(k_W) = 1$? YES K4 default, not invalidated
- (c) k_F scope(D_joint)? YES D_joint demands F's claim be part of joint evaluation

Paper v2.0 4.4 additional conditions:

- (a') k_W is valid registered measurement? YES $_W(M_W)=1$, $V(k_W)=1$
- (b') k_W's content concerns same target as k_F? YES same laboratory F+S
- (c') k_W produced by measurement structurally required YES w's interference measurement to register state of same system k_F registered? measures the lab containing F+S
- (d') No arbitrary privilege? YES only temporal ordering and content incompatibility used

Auth = 1

Check	Status
Auth = 1	All conditions met. k_W has authority over k_F in this C_K.

Step L4-5: Registered contradiction

k_W k_F within C_K?

Check K5 minimal definition:

- k_F registers o_F = |h (definite outcome)
 - k_W registers o_W = |+
 - Can both be treated as valid K-side claims within the same C_K?
- NO |h is a definite state claim; |+ is a superposition that does not preserve |h as a valid claim. Within one C_K, claiming both "outcome is definitely |h" AND "outcome is superposition with no definite |h" is a registration contradiction.

k_W k_F within C_K. [K5 minimal definition satisfied]

Check paper v2.0 4.4 act-level definition:

- Same C_K? YES (L4-3)
- Same registration target? YES (same lab F+S)
- Cannot both satisfy validity conditions? YES (above)
- Later act has valid cross-registration authority? YES (L4-4)

M_W M_F [act-level registered contradiction confirmed]

Check	Status
k_W k_F	Registration contradiction established

Step L4-6: Bridge_EWF

Bridge_EWF(D_joint; M_F, M_W) = ?

Check conditions :

- (a) D_joint requires F-side and W-side registrations to be YES (L4-2) evaluated as jointly valid parts of one lab history?
- (b) M_F registers definite friend-side outcome o_F? YES o_F = |h
- (c) M_W registers same lab as coherent superposition with YES o_W = |+, no definite o_F preserved as W-side valid claim? no |h preserved
- (d) LF/no-go comparison requires both claims to support YES by EWF setup one cross-observer validity constraint?
- (e) No reinterpretation inside same K_joint can preserve CHECK this is the both contents without changing validity of at least relativization one side? defense question

Relativization defense check :

Could K_joint host meta-descriptions ?
 Paper's answer: NO D_joint demands joint validity of original claims, not meta-descriptions. Relativizing abandons D_joint rather than satisfying it.
 This is a FRAMEWORK-LEVEL SEMANTIC COMMITMENT (documented in T3).

Bridge_EWF(D_joint; M_F, M_W) = 1 [assuming relativization defense holds]

Check	Status
Bridge_EWF = 1	All conditions met

Step L4-7: AdmJoint check

Does an admissible $K_joint(K_F, K_W)$ exist?

Candidate $K_joint = i_F(K_F) \quad i_W(K_W) = \{ i_F(k_F), i_W(k_W) \}$

Check AdmJoint conditions :

(i) Embeddings preserve act, outcome, cert, time/order, V?

$i_F(k_F) = M_F, |h, 1, t_F, V_joint(k_F)$

$i_W(k_W) = M_W, |+, 1, t_W, V_joint(k_W)$

Act, outcome, cert preserved? YES.

Order: $t_F < t_W$ in lab history $i_F(k_F) <_{joint} i_W(k_W)$. YES.

V preservation by K8 [$V_X(i(k)) = V_R(k)$ at t_embed]:

$V_joint(i_F(k_F)) = V_F(k_F) = 1$ [k_F native V preserved at embedding]

$V_joint(i_W(k_W)) = V_W(k_W) = 1$ [k_W native V preserved at embedding]

Both $V=1$ carry into K_joint . K8

(ii) Self-certification intrinsic to each embedded act?

$_F(M_F) = 1$ in K_joint ? Must remain intrinsic not redefined by K_joint . YES.

$_W(M_W) = 1$ in K_joint ? Same. YES.

(iii) Conditions 1-6 satisfied for each embedded structure?

For $i_F(k_F)$: C1-C5 carry over. C6 (non-invalidation) CHECK:

Is there k' in K_joint with $k' \quad i_F(k_F)$ and Auth?

$i_W(k_W) \quad i_F(k_F)$ within C_K (established in L4-5)

Auth($i_W(k_W) \quad i_F(k_F)$, C_K) = 1 (established in L4-4)

K5 FIRES: $V(i_F(k_F)) \quad 0$ CONFLICT

For $i_W(k_W)$: C1-C6 no later event contradicts k_W in K_joint . OK.

(iv) No required registration-state update invalidates either embedded structure while both claimed as jointly valid?

K5 just fired: $V(i_F(k_F)) \quad 0$ while both were claimed jointly valid.

AdmJoint condition (iv) VIOLATED. FAILS

$AdmJoint(K_joint; K_F, K_W) = 0$ [no admissible K_joint exists for this model]

Check	Status
AdmJoint = 0	Condition (iv) violated via K5 conflict

Step L4-8: $_K$ conclusion

$K_F \quad _K \quad K_W$?

requires_ $K_joint(F, W) = 1$? YES (L4-1)

admissible K_joint ? NO (L4-7)

$K_F \quad _K \quad K_W$ [K-side incommensurability holds in this model]

Check	Status
$K_F \quad _K \quad K_W$	Incommensurability established for this concrete model

7.4 Consistency Verdict / Kt lun Nht qun

The concrete model is internally consistent. Walking K1-K8 on K_F and K_W individually produces no contradiction. Walking Level 4 definitions on the joint scenario produces a well-defined chain: requires_ $K_joint = 1$ $D_joint = 1$ C_K exists Auth = 1 $k_W \quad k_F$ Bridge_EWF = 1 K5 fires in K_joint AdmJoint(iv) fails $K_F \quad _K \quad K_W$. Each step follows from the previous without circular reasoning within THIS model. The concrete model serves as **evidence of consistency** (a satisfying model exists for all

axioms simultaneously).

Identified gaps (not inconsistencies):

#	Gap	Severity	Location
G1	Relativization defense is framework-level semantic commitment required by this formulation of D_{joint}	Medium	L4-6, step (e). Documented in T3.
G2	K7 closure conditional on D_{joint} resolution	Low	7.2 K7 row. Working as designed.
G3	K5 minimal definition used here; full Level 4 formalization not frozen	Medium	L4-5. Documented in Open Item #14.

7.5 T2 Proof Attempt / Nh p Chng minh T2

Goal: Derive $K_F_K K_W$ from K1-K8 + Level 4 definitions in the concrete model.

Statement to prove:

In the EWF concrete model : if $requires_K_joint(F,W) = 1$ via D_{joint} , and $Bridge_EWF(D_{joint}; M_F, M_W) = 1$, then $K_F_K K_W$.

Proof attempt:

Step 1 Setup :

$K_F = \{k_F\}$ with $k_F = M_F$, $|h, 1, 1, 1$. [K1: well-formed tuple, cert=1]
 $K_W = \{k_W\}$ with $k_W = M_W$, $|+, 1, 2, 1$. [K1: well-formed tuple, cert=1]
 $_F(M_F) = 1$, $_W(M_W) = 1$, independent. [K3: intrinsic self-certification]
 $V(k_F) = 1$, $V(k_W) = 1$ by default. [K4: cert=1 $V=1$, non-null]

Step 2 $requires_K_joint$:

$requires_K_joint(F, W) = 1$.
 Justification: Condition A W performs interference on F+S lab.
 $D_{joint}(K_F, K_W, Arch_EWF) = 1$.
 Source: paper v2.0 4.3 definition. Applied correctly in L4-1, L4-2.
 Confidence: HIGH direct application of sufficient condition A.

Step 3 C_K and Auth :

$C_K(k_F, k_W)$ exists. [L4-3: all three conditions met]
 Auth = 1. [K6 + L4-4: all conditions met]
 Confidence: HIGH mechanical check of conditions.

Step 4 Registered contradiction :

$k_W k_F$ within C_K .
 Justification: $o_F = |h$ (definite), $o_W = |+$.
 These cannot both be valid K-side claims within one C_K .
 Source: K5 minimal definition.
 Confidence: HIGH for K5 minimal. MEDIUM for full Level 4 (not frozen).
 GAP G3: Full Level 4 boundary clauses not frozen. K5 minimal used here is self-contained but may need revision if Level 4 changes semantics.

Step 5 $Bridge_EWF$:

$Bridge_EWF(D_{joint}; M_F, M_W) = 1$.
 Justification: All conditions (a)-(d) mechanically checked in L4-6.
 Condition (e) "no reinterpretation preserves both" depends on

relativization defense .

GAP G1: Relativization defense is a framework-level semantic commitment.

If rejected, Bridge_EWF = 1 does not follow from K1-K8 alone.

Confidence: MEDIUM conditional on this semantic commitment.

Step 6 K5 fires in candidate K_joint :

By K8: embeddings i_F, i_W preserve V values. [K8: $V_X(i(k)) = V_R(k)$]

In candidate K_joint:

$i_F(k_F) <_{\text{joint}} i_W(k_W)$ [K2: $t_F < t_W$]

$i_W(k_W) \ i_F(k_F)$ within C_K [Step 4]

$\text{Auth}(i_W(k_W) \ i_F(k_F), C_K) = 1$ [Step 3]

K5: $V_{\text{prov}}(i_F(k_F)) \neq \emptyset$ [K5 pre-closure invalidation]

At this stage, V_{final} has not yet been assigned: K7 closure can occur only after the pending requires_K_joint demand is resolved.

This happens while D_{joint} claims both as jointly valid.

AdmJoint condition (iv) violated.

Confidence: HIGH direct K5 + K8 application. No gap.

Step 7 Conclusion :

$\text{requires_K_joint}(F, W) = 1$ [Step 2]

$\text{K_joint: AdmJoint}(K_{\text{joint}}; K_F, K_W) = 1$ [Step 6]

$K_F _K \ K_W$ [Definition of $_K$, paper v2.0 4.4]

(conditional)

7.6 Proof Attempt Assessment

Step	Confidence	Depends on	Gap?
1 (Setup)	HIGH	K1, K3, K4, K8	None
2 (requires_K_joint)	HIGH	Level 4 4.3 Condition A definition	Level 4 not frozen
3 (C_K, Auth)	HIGH	K6 + Level 4 4.4	Level 4 not frozen
4	HIGH/MEDIUM	K5 minimal / Level 4 full	G3 : Level 4 not frozen
5 (Bridge_EWF)	MEDIUM	External philosophical assumption	G1 : Relativization defense
6 (K5 in K_joint)	HIGH	K5 + K8	None (resolved by K8)
7 (Conclusion)	HIGH	Steps 2+6 + $_K$ definition	Level 4 definition

Overall assessment:

The proof attempt is **valid conditional on two remaining identified dependencies** (G1, G3). Neither is an internal contradiction one is a philosophical boundary, one is a temporal dependency: - **G1 (Relativization defense)**: Framework-level semantic commitment not derived within K1-K8. This formulation makes explicit what counts as "satisfying a joint validity demand." Documented as a semantic boundary, not a mathematical gap. - **G3** : Temporal dependency resolves when paper v2.0 Level 4 boundary clauses are frozen. K5 minimal is sufficient for the concrete model. **Former EP gap (G1 in v1.3): RESOLVED**. EP promoted to K8 (v1.4) V-preservation through cross-space embedding is now a core axiom. The proof chain no longer depends on an external postulate for Step 6. The circularity concern from v1.2 Open Item #14 is **not present in this concrete model** because K5's minimal definition is sufficient for Step 4 without invoking Level 4's full formalization. The circularity only appears in the GENERAL case where T2 needs AdmJoint conditions that reference full . In the concrete model, is directly verified by content inspection .

7.7 Next Steps / Bc Tip theo

Following the 5-step methodology:

Step	Status	Timeline estimate
Step 1 Concrete Model	DONE consistency established	
Step 2 Proof attempt for T2	DONE 2 remaining dependencies . Former EP gap resolved by K8.	
Step 3 Submit K-Axiom + Concrete Model to PhilSci	Ready for community review	1-2 weeks
Step 4 Based on feedback, decide: close remaining gaps or find collaborator	Pending feedback	After Step 3
Extension Generalize from N=2 to N>2	T4 verification (Open Item #9)	After Step 4

Decision point after community feedback:

- If gaps G1, G3 are accepted as documented Level 4 freeze proceeds
- If G1 (relativization) is challenged T3 needs revision, but K1-K8 unchanged (this is a philosophical/semantic challenge, not mathematical)
- If consistency check reveals new issues return to concrete model, extend

8. Open Items / Cc mc M

#	Item	Status	Priority
1	Multi-step retroactive chain (E8 extension)	Deferred K5 single-step; K5 V_prov pre-closure mechanism allows re-assessment of invalidating acts before K7 closure . Multi-step chain requires additional axiom(s).	Medium
2	Null K-state full formalization (E9 detailed operationalization)	Partial K1 o= + K4 E9 exception structurally accommodate null events. Detailed operationalization deferred.	Low-Medium
3	Validated absence validity conditions (E14 extension)	Partial K1 o= + K4 default validity structurally accommodate. Specific absence validity conditions deferred.	Medium
4	Inter-K-space relation structure (E15 extension)	Deferred new axiom needed	Low-Medium

#	Item	Status	Priority
5	Pre-registration K-state (E16 extension / K0)	Deferred new axiom needed	Low-Medium
6	Pre-symbolic registration stratum (E4 formalization)	Deferred K-space stratification	Low
7	Equivalence of (M) and R_svasa formalisms	Deferred separate research track	Low
8	Full semantic proof for Bridge_EWF "no admissible reinterpretation"	Pending Level 4 freeze + T3 completion. External assumption (relativization defense) documented.	High
9	T4 N>2 verification	Requires multi-observer EWF modeling	Medium
10	Update paper v2.0 Section 7.2 deferred item #5 status	After community feedback on this document	Low
11	RCA re-audit after community feedback	After Level 4 freeze and T1-T3 finalization	High
12	K6 Auth non-transitivity edge cases (circular authority chains)	Resolved v1.2 counterexample provided in K6 formal block. Remaining: N3 exotic topologies.	Low
13	Embedding Postulate (EP) promotion decision	Resolved v1.4 EP promoted to K8 (Cross-Space Embedding Preservation). K8 is now a frozen Layer 1 core axiom. T1-T3 no longer depend on an external postulate for V-preservation.	~~High~~ Resolved

#	Item	Status	Priority
14	T2 circularity resolution Level 4 freeze	T2 derivation is conditional on Level 4 formalization being consistent with K5 minimal definition. Circularity resolves when Level 4 freezes. v1.3 update: Circularity NOT present in concrete model K5 minimal is directly verifiable by content inspection. Circularity remains only in general case. v1.4/Phase 2 update: T2 also documented as K7 Dep-B (F6b + F7b): T2's AdmJoint(iv) operates on V_prov during pre-closure admissibility testing; resolved-demand outcomes supply K7 closure semantics. This is a Layer 2 (updatable) dependency K1-K8 unchanged.	High
15	Concrete model gaps G1-G3	G1 (Relativization): framework-level semantic commitment required by this formulation of D_joint. G2 (K7 closure): working as designed. G3 : see #14. All gaps are external dependencies, not internal contradictions. v1.4: Former EP gap resolved by K8. Renumbered G1-G4 G1-G3. Phase 2 note: Dep-A and Dep-B (T1 <_joint> ordering from K2+K8) are satisfied dependencies in the concrete model not open gaps. Documented in K5/K6/K7 Dependency rows.	Medium

9. Cross-References / Tham chiu Cho

Document	Relationship
papers/.../VVV-QMRF_Working_Paper_v2.0.md	Downstream K-space axioms are the foundation for 3-4 structural definitions

Document	Relationship
meta_architecture/vvv_qmrf_meta_architecture_registration_layer_formalization.md	Upstream Defines the K-state tuple, (M), V(M), and the symbol registry that K1-K5 axiomatize
framework/vvv_qmrf_framework_e01_self_certifying_registration_postulate.md	Upstream Source for K3
framework/vvv_qmrf_framework_e06_registering_system_as_process_postulate.md	Upstream Source for K2
framework/vvv_qmrf_framework_e07_registration_validity_location_postulate.md	Upstream Source for K4, K5 (<i>internal project reference</i>)
synthesis/vvv_qmrf_synthesis_s3_registering_system_as_process_foundation.md	Upstream Source for K2 discreteness (<i>internal project reference</i>)
SYSTEM_Buddhist_Epistemology/system_be_full.md	Diagonal BE SOT for lineage annotations (<i>internal project reference</i>)
vvv-qmrf/schema_guide.md	Process Document creation contract (<i>internal project reference</i>)

10. Level 4 Freeze Check Internal Consistency Verdict / Phn quy t Nht qun Ni ti

10.1 Question / Cu hi

Can the Level 4 definitions be formally proven consistent with K1-K8 without external confirmation?

10.2 RCA Trace / Truy vt RCA

Step	Question	Answer		
Define	What is the "Level 4 freeze check"?	A formal proof that all Level 4 definitions from paper v2.0 4.3-4.5 are consistent with Layer 1 axioms K1-K8 no contradiction arises when combining them.		
Trace (Why 1)	Why is this needed?	Level 4 is currently the least stable layer (in community review). Before freezing it, internal consistency must be established so that community feedback does not hit contradictions that could have been caught internally.		

Step	Question	Answer		
Trace (Why 2)	Why is it hard?	Because Level 4 definitions were designed bottom-up (from EWF use case) while K1-K8 were designed top-down (from BE structural sources). The two directions may not converge. Plus, Level 4 uses complex primitives whose full formalization is not yet frozen.		
Trace (Why 3)	Why can't it be purely internal?	Because one Level 4 dependency the relativization defense (Bridge_EWF condition (e), paper v2.0 4.5) is a SEMANTIC choice about what counts as "satisfying D_joint." This document treats that boundary as external to K1-K8: VVV-QMRF makes the stance explicit through the relativization defense, rather than deriving it from the K-space axioms themselves.		
Isolate	What are the blockers?	(1) Relativization defense = framework-level semantic commitment required by this formulation of D_joint (not a mathematical gap). (2) Full Level 4 boundary clauses = not yet frozen (temporal dependency). (3) General case proof (arbitrary	K_R	and N observers) = requires stronger mathematical foundations.

Step	Question	Answer		
Fix cause	What CAN be done internally?	Promote EP K8 (DONE v1.4). Concrete model consistency proof (DONE v1.3). Edge case testing. General case proof sketch with explicit boundary documentation.		
Verify	How to verify?	Walk every Level 4 definition against K1-K8 in the concrete model . Check no contradiction arises. Document what IS proven vs. what depends on external assumptions.		

10.3 What CAN Be Proven Internally / C th Chng minh Ni ti

#	Statement	Status	Confidence
P1	K1-K8 are internally consistent (concrete model: 2 observers, 1 event each)	PROVEN	HIGH
P2	Level 4 definitions can be expressed in terms of K1-K8 primitives	PROVEN	HIGH
P3	The derivation chain requires_K_joint D_joint C_K Auth Bridge_EWF K5 fires AdmJoint fails _K is well-defined (no circular reasoning)	PROVEN	HIGH
P4	Step 6 (K5 fires in K_joint) does NOT depend on any external postulate	PROVEN	HIGH
P5	K5 minimal can be verified by content inspection without invoking Level 4 full (circularity absent in concrete model)	PROVEN	HIGH
P6	K_joint candidate existence is derivable from K1-K8 + Level 4 scope identifiers	PROVEN (T1, updated v1.4)	HIGH

10.4 What CANNOT Be Proven Internally / Khng th Chng minh Ni ti

#	Statement	Why not	Nature of boundary		
E1	Relativization defense: "meta-descriptions do not satisfy D_{joint} "	Semantic choice about the nature of joint validity not derived within K1-K8	Framework-level semantic commitment required by this formulation of D_{joint}		
E2	Full Level 4 boundary clauses are correct	Still in community review (paper v2.0). Internal consistency with K5 minimal can be checked, but community may disagree with boundary clauses.	Temporal dependency (resolves when Level 4 freezes)		
E3	General case proof (arbitrary N, arbitrary)	K_R	)	Requires stronger mathematical foundations (structural induction proofs, category-theoretic colimit properties for $N > 2$).	Mathematical capacity boundary (documented honestly)

10.5 Final Verdict / Phn quyt Cui cng

Level 4 definitions ARE internally consistent with K1-K8 in the concrete model. The proof chain has exactly **one declared external dependency**: the relativization defense (a framework-level semantic commitment about what counts as satisfying D_{joint} in this formulation). This is NOT an internal contradiction it is a **semantic boundary** that VVV-QMRF declares explicitly rather than deriving from K1-K8. Copenhagen, Many-Worlds, QBism, and VVV-QMRF make different semantic choices about joint validity; this document only commits VVV-QMRF to the relativization defense. **Decision:** Level 4 can freeze with **one documented framework-level semantic commitment** (relativization defense). The former EP gap (G1 in v1.3) is resolved by promoting EP K8. The circularity concern (Open Item #14) is absent in the concrete model. The remaining temporal dependency resolves when paper v2.0 exits community review. **Confidence level for Level 4 freeze: MEDIUM-HIGH.** Internal consistency is proven for the relevant model class. The remaining blocker is the framework-level semantic commitment about what counts as satisfying D_{joint} in this formulation.

10.6 Remaining Action Items After Freeze / Cc Mc Hnh ng Sau Freeze

#	Item	Priority	Blocks		
A1	Document relativization defense as "Axiom of Joint Validity Semantics" (separate from K1-K8)	High	T3 completeness		
A2	Freeze Level 4 boundary clauses after community feedback resolves Open Item #14	High	T2 non-circularity in general case		
A3	General case proof (structural induction on	K_R	, N observers)	Medium	T4, E8, E15
A4	Edge case: E9 null events, E14 validated absence	Medium	E8-E16 audit phase		
A5	Category-theoretic formalization of K_{joint} as colimit ($N > 2$)	Low-Medium	T4		
A6	When Level 4 freezes, verify that conditional semantic dependencies Dep-A and Dep-B ($T1 <_{\text{joint}} >$ ordering via $K2+K8+\text{Level } 4$) documented in K5/K6/K7 Dependency rows remain consistent with frozen Level 4 extensional definitions	Medium	Level 4 freeze (resolves Open Item #14)		

Document v1.5 2026-05-19 VVV-QMRF K-AXIOM Status: Class D (proposed). All axioms and theorems are proposed registration-layer definitions. Layer 1 (K1-K8): Frozen. Layer 2 (T1-T4): Updatable pending Level 4 freeze. RCA audit : Full Phase 15 RCA audit completed (plan v28). Phase 1 : K5 V_prov/V_final lifecycle split (F1, BLOCKING resolved); K6 non-transitivity scoped to distinct C_K contexts (F2); 0.5 isolation paragraph

2-part split (F3); Layer 1 Summary C_K roles (F4); K5 K_R disambiguation + firing precondition + Dep-A/Dep-B documented . Phase 2 : K6/K7 Dep-A (C_K precondition) + I-03 pattern documented ; C-KAXIOM-010 rewritten as 2-part syntactic/semantic isolation (F6c). Phase 3 : T1 non-circularity guard (F7a); T2 AdmJoint V_prov timing + K7 resolved-demand semantics (F7b); T3 framework-level semantic boundary wording (F7c); T4 global commutativity guard (F7d). Phase 4 : E2 K1 vs K4/K7 boundary; E9 definitional null-status boundary; E8 V_prov/T2/E9 precision; BE lineage expanded to 8/8 PASS . Phase 5 : 7.5 Step 6 V_prov notation (F9a); 7.5 Step 4 stale GAP G4 G3 label (F9b); 10.3 P4 citation V_prov internal note + 7.5 Step 6 stale "modulo EP" removed (F9c); 7.3 L4-7 K8 canonical V_F/V_W subscript notation (F9d). Phase 6 : Open Item #1 K5 V_prov attribution (F10a); Open Item #14 T2 Dep-B note (F10b); Open Item #15 Dep-A/Dep-B satisfied note (F10c); Action Item A6 added Dep-A/Dep-B post-freeze verification (F10d); document header and version history updated . RCA audit : (1) EP promoted to K8 (Cross-Space Embedding Preservation) Layer 1 now has 8 core axioms. K8 guarantees V-preservation through cross-space embeddings. (2) T1 derivation updated: V-preservation now from K8, not external postulate. Former EP gap (G1) RESOLVED. (3) T2 proof attempt gaps reduced from 3 to 2: only relativization defense (G1, framework-level semantic commitment) and Level 4 freeze (G3, temporal) remain. (4) Concrete model 7 updated: K8 consistency walk, AdmJoint check (i) now derives from K8. (5) 10 Level 4 Freeze Check verdict added: internal consistency PROVEN for concrete model; relativization defense documented as framework-level semantic boundary. (6) Open Item #13 closed . Open Items #14, #15 updated. Previous : (1) Concrete model 7 added: minimal EWF (2 observers, 1 event each). K1-K7 consistency walk completed no contradictions. Level 4 definitions walk completed derivation chain verified. (2) T2 proof attempt with 3 gaps. (3) Circularity shown absent in concrete model. (4) Open Items #14, #15 added. Previous : K2 corrected to total order. T1 EP gap acknowledged. K6 non-transitivity counterexample. T2 circularity acknowledgment. Previous : Added K6, K7, K4 E9 exception, K5 minimal definition, K1 cert admission rule. Fixed T1 V-preservation (EP), T2 sufficient-vs-necessary, T3 external assumption. Companion: VVV-QMRF Working Paper v2.0 published on Zenodo (2026-05-19) <https://doi.org/10.5281/zenodo.20289261> Next: Community feedback Level 4 boundary clauses freeze (resolves #14) T1-T3 finalization N>2 generalization (T4, #9) E8-E16 extension audit phase.